Accoppiamento FEM/BEM: implementazione delle condizioni di dominio aperto nel codice FEM 2D elettrostatico *

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ABSTRACT

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Key words: Keyword1, Keyword2, Keyword3

1 INTRODUCTION

A classic example of FEM/BEM coupling in an electrostatic problem is the computation of the electric field generated by a parallel-plate capacitor. In a simulation of this type, it is often useful to visualize not only the behavior of the field inside the capacitor, but also outside it. In Physics II courses, the external field is usually assumed to be zero; in reality, due to edge effects, it is not negligible, and in some applications it can play a significant role. This is the main motivation for analyzing in greater detail the behavior of the electric field in such a problem.

In this project, we consider the relatively simple case of a parallel-plate capacitor. Besides the charges present on the plates, no other sources are considered in the surrounding space. The differential equation to be solved is Maxwell's first law (1) in its simplified electrostatic form (2), expressed by treating the electric field as the gradient of the potential (3).

$$\nabla \cdot D = \rho \tag{1}$$

$$\nabla \cdot E = 0 \tag{2}$$

$$\Delta V = 0 \tag{3}$$

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A well-known first approach consists in using the Finite Element Method (FEM). One defines a domain large enough to contain the capacitor and imposes a zero potential on its boundary, assuming that at a sufficiently large distance the potential effectively tends to zero. However, this method has some drawbacks: it requires very large domains and therefore large meshes, resulting in increased computational time. Moreover, the mesh resolution must be uniform inside and outside the capacitor, even though the quality required to obtain an accurate result is not the same in the two regions. Computing the external field with FEM therefore leads to significant computational waste, and in any case the potential cannot be represented up to infinity, introducing an unavoidable approximation.

The Boundary Element Method (BEM), on the other hand, offers a more efficient solution to the problem. By reducing the dimension of the domain by one order, it allows substantial computational advantages. By its nature, BEM enables the accurate computation of the unknown variable at any internal point of a domain, solving the problem only on its boundary (fig.). BEM is also applicable to very large or unbounded domains (fig.), provided that the asymptotic behavior of the potential is known a priori and that it converges as $1/r^2$. In such cases, BEM can correctly represent the behavior at infinity without introducing artificial approximations.

The goal of this work is therefore to combine FEM and BEM to obtain more accurate results at lower computational cost. The idea is to use FEM to determine the solution accurately inside the capacitor and employ BEM to extend it outward to infinity in an exact manner. The coupling mechanism between the two methods will be described in detail in the dedicated section.

2 COUPLED

2.1 FEM formulation

In this section we derive the FEM formulation for our problem, we do not perform the FEM as usual, since we need to achieve a specific formulation in order to couple FEM and BEM.

The problem to solve is (4)

$$\Delta u = 0 \quad \text{in } \Omega. \tag{4}$$

Let's rewrite the problem in the weak formulation (5)

$$\text{Find } u \in H^1(\Omega) \quad \text{s.t} \quad \int_\Omega \nabla u \cdot \nabla v \, dx - \int_{\partial\Omega} \frac{\partial u}{dn} v \, ds = 0 \qquad \forall v \in H^1_g(\Omega) := \{v \in H^1 \text{ s.t. } v...\} \tag{5}$$

We did not write boundary conditions for the problem. this is because we have neither Dirichlet nor Neumann boundary conditions; we have a relation between u and $\frac{\partial u}{dn}$ thanks to the BEM method. So, for now, let's write the problem in this way without imposing any bcs.

Now we consider that to have equation (5) verified $\forall v$ it's sufficient to verify for each element of the base of H^1. We also discretize the problem and so do not solve the problem for $v \in H^1$ but for $v_h \in V_h \subset H^1$. These consideration leads to N_h equations, where N_h is the dimension of the space V_h. Now we observe that $u \in V_h$ therefore we can write u as $\sum_{j=1}^{N_h} u_j v_j$ where v_j are the basis of V_h. We can now write the problem as in (6) where v_i forms the base of the V_h space. We can see (6) as a system of N_h equations in N_h unknowns which are u_j. If we achieve u_h we can perform $\sum_{j=1}^{N_h} u_j v_j$ and obtain $u \in V_h$ which is the solution of our problem.

$$\text{Find } u_j \text{ , } j = 1, ..., N_h \quad \text{s.t} \quad \int_\Omega \nabla(\sum_{j=1}^{N_h} u_j v_j) \cdot \nabla v_i \, dx - \int_{\partial\Omega} \frac{\partial(\sum_{j=1}^{N_h} u_j v_j)}{dn} v_i \, ds = 0 \qquad \forall v_i \text{ , } i = 1, ..., N_h \tag{6}$$

With this notation we lose information about the normal derivative since we can rearrange the term and take out u_j that are constant from the derivative. So to mantain a $\frac{\partial u}{dn}$ we consider that $\frac{\partial u}{dn}$ can be written as $\sum_{j=1}^{N_h} x_j v_j$ where v_j are the basis of V_h and x_j are coefficient. In fact since we can select the space V_h, it's possibile to choose a space who garantee that both u and $\frac{\partial u}{dn}$ are element of this space an therefore written as a proper linear combination of the basis of the space. We rewrite (6) as (7), in the new formulation x_i represents the coefficient to find to achieve the normal derivative. To highlight the meaning of x_i we report also (8) in which x_i are called $\frac{\partial u_i}{dn}$, is important to pay attention to this notation because we are not appling the normal derivative operator to u_i which is a coefficient, and the operator would not be meaningful since the operation would result in 0. We are just denoting the coefficient with the symbol $\frac{\partial u_i}{dn}$

$$\text{Find } u_j, x_j \text{ , } j = 1, ..., N_h \quad \text{s.t} \quad \int_\Omega \nabla(\sum_{j=1}^{N_h} u_j v_j) \cdot \nabla v_i \, dx \; - \int_{\partial\Omega} (\sum_{j=1}^{N_h} x_j v_j) v_i \, ds = 0 \qquad \forall v_i \text{ , } i = 1, ..., N_h \tag{7}$$

$$\text{Find } u_j, x_j \text{ , } j = 1, ..., N_h \quad \text{s.t} \quad \int_\Omega \nabla(\sum_{j=1}^{N_h} u_j v_j) \cdot \nabla v_i \, dx \; - \int_{\partial\Omega} (\sum_{j=1}^{N_h} \frac{\partial u_j}{dn} v_j) v_i \, ds = 0 \qquad \forall v_i \text{ , } i = 1, ..., N_h \tag{8}$$

$$\text{Find } u_j, x_j \text{ , } j = 1, ..., N_h \quad \text{s.t} \quad \int_\Omega \sum_{j=1}^{N_h} (u_j \nabla v_j \cdot \nabla v_i) \, dx \; - \int_{\partial\Omega} (\sum_{j=1}^{N_h} \frac{\partial u_j}{dn} v_j v_i) \, ds = 0 \qquad \forall v_i \text{ , } i = 1, ..., N_h \tag{9}$$

$$\text{Find } u_j, x_j \text{ , } j = 1, ..., N_h \quad \text{s.t} \quad \sum_{j=1}^{N_h} (u_j \int_\Omega \nabla v_j \cdot \nabla v_i \, dx) \; - (\sum_{j=1}^{N_h} \frac{\partial u_j}{dn} \int_{\partial\Omega} v_j v_i \, ds) = 0 \qquad \forall v_i \text{ , } i = 1, ..., N_h \tag{10}$$

Doing some transformation we can rewrite the problem as (11)

$$\text{Find } u, \frac{\partial u}{\partial n} \text{ s.t.} \quad S_i \, u - T_i \frac{\partial u}{\partial n} = 0 \qquad \forall v_i, \quad i = 1, \ldots, N_h \quad \text{where:} \tag{11}$$

$$A_i = \left[\int_\Omega \nabla v_1 \cdot \nabla v_i \quad \int_\Omega \nabla v_2 \cdot \nabla v_i \quad \ldots \quad \int_\Omega \nabla v_{N_h} \cdot \nabla v_i \right] \tag{12}$$

$$\mathbf{u} = \begin{bmatrix} u_1 & u_2 & \ldots & u_{N_h} \end{bmatrix}^T, \tag{13}$$

$$T_i = \left[\int_{\partial\Omega} v_1 v_i \quad \int_{\partial\Omega} v_2 v_i \quad \ldots \quad \int_{\partial\Omega} v_{N_h} v_i \quad \right] \tag{14}$$

$$\frac{\partial \mathbf{u}}{\partial n} = \begin{bmatrix} \frac{\partial u_1}{\partial n} & \frac{\partial u_2}{\partial n} & \ldots & \frac{\partial u_{N_h}}{\partial n} \end{bmatrix}^T. \tag{15}$$

Visualize the N_h equation as a system we can write (16)

$$\text{Find } u, \frac{\partial u}{\partial n} \quad \text{s.t} \quad S u - T \frac{\partial u}{dn} = 0 \tag{16}$$

2.2 BEM formulation

In this section we derive the BEM formulation for our problem. The key idea behind BEM is to reformulate the problem defined on Ω as an equivalent problem defined only on the boundary $\partial\Omega$, reducing the dimensionality of the problem.

The starting point is the fundamental solution of the Laplace operator. We define $G(x, y)$ as the function satisfying

$$\Delta_x G(x, y) = -\delta(x - y) \quad \text{in } \mathbb{R}^d, \tag{17}$$

where δ is the Dirac delta distribution and d is the space dimension. The explicit expression of G is:

$$G(x, y) = \begin{cases} -\dfrac{1}{2\pi} \log |x - y| & \text{if } d = 2, \\ \dfrac{1}{4\pi|x - y|} & \text{if } d = 3. \end{cases} \tag{18}$$

2.2.1 Green's third identity

Let $u \in H^1(\Omega)$ be a solution of (4). Applying Green's second identity to u and $G(\cdot, y)$ for a fixed $y \in \Omega$ yields the *representation formula* (19):

$$u(y) = \int_{\partial\Omega} G(x, y) \frac{\partial u}{\partial n}(x)\, ds_x - \int_{\partial\Omega} \frac{\partial G}{\partial n_x}(x, y)\, u(x)\, ds_x, \qquad \forall y \in \Omega. \tag{19}$$

This formula expresses u at any interior point y solely in terms of the boundary values of u and $\frac{\partial u}{\partial n}$. We define the two integral operators appearing in (19):

$$(\mathcal{V}\phi)(y) := \int_{\partial\Omega} G(x, y)\, \phi(x)\, ds_x, \tag{20}$$

$$(\mathcal{K}\phi)(y) := \int_{\partial\Omega} \frac{\partial G}{\partial n_x}(x, y)\, \phi(x)\, ds_x, \tag{21}$$

where $\mathcal{V}$ is called the *single layer operator* and $\mathcal{K}$ is called the *double layer operator*. With this notation, (19) becomes:

$$u(y) = \mathcal{V}\left(\frac{\partial u}{\partial n}\right)(y) - \mathcal{K}(u)(y), \qquad \forall y \in \Omega. \tag{22}$$

2.2.2 Boundary Integral Equation

To obtain a problem posed exclusively on $\partial\Omega$, we take the trace of the representation formula, i.e., we let $y \to \partial\Omega$. Due to the singularity of G, this limit must be handled carefully and leads to the *boundary integral equation* (BIE):

$$\frac{1}{2} u(y) = \mathcal{V}\left(\frac{\partial u}{\partial n}\right)(y) - \mathcal{K}(u)(y), \qquad \forall y \in \partial\Omega. \tag{23}$$

The factor $\frac{1}{2}$ arises from the jump relations of the double layer potential across a **smooth** boundary. Equation (23) is a relation between $u|_{\partial\Omega}$ and $\frac{\partial u}{\partial n}|_{\partial\Omega}$: given one of the two, one can in principle recover the other. *Remember that the factor $\frac{1}{2}$ is in general a value α, depends on α the equation is valid for inside the domain, on a smooth border, on a border with corners or outside of the domain. In general $\alpha = \frac{\theta}{2\pi}$*

2.2.3 *Discretization*

We now discretize (23). Let $V_h \subset H^{1/2}(\partial\Omega)$ be a finite-dimensional space of dimension N_h^∂, with basis $\{\psi_j\}_{j=1}^{N_h^\partial}$. We approximate the boundary unknowns as:

$$u_h = \sum_{j=1}^{N_h^\partial} u_j\, \psi_j, \qquad \frac{\partial u_h}{\partial n} = \sum_{j=1}^{N_h^\partial} \frac{\partial u_j}{\partial n}\, \psi_j. \tag{24}$$

Following the same notation introduced for FEM, **we remark that $\frac{\partial u_j}{\partial n}$ denotes a coefficient**, not the application of the normal derivative operator to the scalar u_j. Substituting (24) into (23) and testing against ψ_i we obtain the discrete system:

$$\frac{1}{2}\sum_{j=1}^{N_h^\partial} u_j \int_{\partial\Omega} \psi_j \psi_i\, ds = \sum_{j=1}^{N_h^\partial} \frac{\partial u_j}{\partial n} \int_{\partial\Omega} \mathcal{V}(\psi_j)\, \psi_i\, ds - \sum_{j=1}^{N_h^\partial} u_j \int_{\partial\Omega} \mathcal{K}(\psi_j)\, \psi_i\, ds, \qquad \forall i = 1, \dots, N_h^\partial. \tag{25}$$

We define the matrices:

$$M_{ij} = \int_{\partial\Omega} \psi_j\, \psi_i\, ds, \tag{26}$$

$$G_{ij} = \int_{\partial\Omega} \mathcal{V}(\psi_j)\, \psi_i\, ds, \tag{27}$$

$$K_{ij} = \int_{\partial\Omega} \mathcal{K}(\psi_j)\, \psi_i\, ds. \tag{28}$$

With this notation, (25) can be written compactly as:

$$\text{Find } u, \frac{\partial u}{\partial n} \quad \text{s.t.} \quad \left(\frac{1}{2}M + K\right) u = G\, \frac{\partial u}{\partial n}. \tag{29}$$

The problem (29) can be written as:

$$\text{Find } u, \frac{\partial u}{\partial n} \quad \text{s.t.} \quad H u = G\, \frac{\partial u}{\partial n}. \tag{30}$$

We remark that problem (30) **is valid to find $u, \frac{\partial u}{\partial n}$ on the border of the domain Ω**

2.3 FEM-BEM Coupling

Before combining the two formulations, it is important to observe their different nature. The FEM system (16) consists of N_h equations and involves all nodal values of u together with the normal derivative coefficients $\frac{\partial u}{\partial n}$, which are defined only on $\partial\Omega$. If we denote by N_h^∂ the number of boundary nodes and split the degrees of freedom accordingly, the total number of unknowns is $N_h + N_h^\partial$, while FEM alone provides only N_h equations. The system is therefore underdetermined: the FEM formulation cannot close without additional information on $\frac{\partial u}{\partial n}$.

It is instructive to look at the FEM equations node by node. For an *interior* node i, the base function $v_i|_{\partial\Omega} = 0$, the boundary term $T_i \frac{\partial u}{\partial n}$ vanishes identically, **this is *not* a general property of H^1 bases, but rather a consequence of a specific choice of basis.** For instance, if one uses *Lagrange nodal basis functions*, then by construction $v_i(x_j) = \delta_{ij}$; for an interior node i, all boundary nodes $x_j \in \partial\Omega$ satisfy

$v_i(x_j) = 0$. Therefore the equation reduces to:

$$S_i\, u = 0, \tag{31}$$

which is already closed in the unknowns u. For a *boundary* node i, the term $T_i \frac{\partial u}{\partial n} \neq 0$ in general, so the equation reads:

$$S_i\, u - T_i\, \frac{\partial u}{\partial n} = 0, \tag{32}$$

and involves the additional unknown $\frac{\partial u}{\partial n}$. These N_h^∂ equations are the ones that remain open. This is precisely where BEM enters. The BEM system (30) provides exactly N_h^∂ equations, but crucially only involves quantities defined on $\partial\Omega$: it is a relation between $u|_{\partial\Omega}$ and $\frac{\partial u}{\partial n}|_{\partial\Omega}$. The coupling strategy is therefore the following: *for boundary nodes, replace the open FEM equations* (32) *with the BEM equations* (30). This is conceptually analogous to imposing Dirichlet conditions in a standard FEM problem: in that case some rows of the system matrix are replaced by $u_j = g_j$; here, the BEM equation plays the same role for the boundary rows, providing the information that the FEM formulation alone cannot supply.

After the substitution, the total system has $N_h + N_h^\partial$ equations in $N_h + N_h^\partial$ unknowns and takes the form (33):

$$\begin{bmatrix} S_{II} & S_{IB} & 0 \\ S_{BI} & S_{BB} & -T_{BB} \\ 0 & H & -G \end{bmatrix} \begin{bmatrix} u_I \\ u_B \\ \dfrac{\partial u_B}{\partial n} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \tag{33}$$

where the subscripts I and B denote interior and boundary degrees of freedom respectively, S_{II} and S_{IB}, S_{BI}, S_{BB} are the blocks of the FEM stiffness matrix S partitioned accordingly, T_{BB} is the boundary mass block from the FEM formulation, and M, K, V are the BEM matrices defined in (30).

The block structure of (33) makes the role of each component transparent: the first block of rows, coming entirely from FEM, governs the interior of the domain and does not involve $\frac{\partial u}{\partial n}$; the second block of rows, also from FEM, couples interior and boundary values of u with the normal derivative; the third block of rows, from BEM, closes the system by providing the missing relation between u_B and $\frac{\partial u_B}{\partial n}$ through the integral operators on $\partial\Omega$.

2.3.1 Remarks on the block structure

The blocks S_{IB} and S_{BI}, are subblocks of the FEM stiffness matrix, whose entries are $S_{ij} = \int_\Omega \nabla v_i \cdot \nabla v_j\, dx$. Since this integral is over the *whole domain* Ω and not just the boundary, it is in general nonzero even when one index is interior and the other is on the boundary: if the supports of v_i and v_j overlap near $\partial\Omega$, the integral is nonzero. Therefore $S_{IB} \neq 0$ and $S_{BI} \neq 0$ in general.

2.3.2 Reduced system after elimination

We can proceed by analogy of imposing dirichlet condition and go for elimination, **we assume G is invertible**:

$$\frac{\partial u_B}{\partial n} = G^{-1} H u_B. \tag{34}$$

Substituting (34) into the second block row of (33):

$$S_{BI}\, u_I + S_{BB}\, u_B - T_{BB}\, G^{-1} H u_B = 0, \tag{35}$$

In this way we eliminate the third block row and the reduced system have only the unknowns u_I and u_B:

$$\begin{bmatrix} S_{II} & S_{IB} \\ S_{BI} & S_{BB} - T_{BB}\, G^{-1} H \end{bmatrix} \begin{bmatrix} u_I \\ u_B \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \tag{36}$$

The matrix in (36) has size $N_h \times N_h$ and is the sum of the standard FEM stiffness contribution and a correction term $T_{BB}\, G^{-1} H$ that is dense and restricted to boundary degrees of freedom, reflecting the nonlocal nature of the BEM operator. Once u_I and u_B are obtained by solving (36), the normal derivative can be recovered a posteriori via (34).

2.3.3 Matrix entries

For convenience of implementation, we collect here the explicit expression of the generic (i, j) entry of each matrix appearing in (36).

The FEM stiffness matrix S has entries:

$$S_{ij} = \int_\Omega \nabla v_i \cdot \nabla v_j \, dx, \tag{37}$$

where $v_i, v_j \in V_h$ range over all N_h basis functions of the domain discretization.

The FEM boundary mass matrix T has entries:

$$T_{ij} = \int_{\partial\Omega} v_i \, v_j \, ds, \tag{38}$$

where in the block notation of (33) only the subblock T_{BB} is relevant, with i, j ranging over boundary nodes only.

The BEM single layer matrix G has entries:

$$G_{ij} = \int_{\partial\Omega} \mathcal{V}(\psi_j)(y)\, \psi_i(y)\, ds_y = \int_{\partial\Omega}\int_{\partial\Omega} G(x, y)\, \psi_j(x)\, \psi_i(y)\, ds_x\, ds_y, \tag{39}$$

where $\psi_i, \psi_j \in V_h^\partial$ are the basis functions of the boundary discretization and $G(x, y)$ is the fundamental solution (18). Note that G_{ij} involves a double boundary integral and is symmetric.

The BEM double layer matrix H has entries:

$$H_{ij} = \frac{1}{2}\int_{\partial\Omega} \psi_i\, \psi_j\, ds + \int_{\partial\Omega} \mathcal{K}(\psi_j)(y)\, \psi_i(y)\, ds_y = \frac{1}{2} M_{ij} + \int_{\partial\Omega}\int_{\partial\Omega} \frac{\partial G}{\partial n_x}(x, y)\, \psi_j(x)\, \psi_i(y)\, ds_x\, ds_y, \tag{40}$$

where the first term is the scaled boundary mass contribution and the second involves the normal derivative of the fundamental solution with respect to the source point x. Unlike G, the matrix H is in general non-symmetric due to the directional nature of the double layer kernel.

3 USING FEM-BEM COUPLED IN THE PROBLEM

3.1 Application to the parallel plate capacitor

3.1.1 Geometry and problem setup

We consider two infinitely thin horizontal plates in 2D, placed at $y = +d/2$ and $y = -d/2$, extending from $x = -L/2$ to $x = +L/2$. The problem is $\Delta u = 0$ in $\mathbb{R}^2 \setminus \{\text{plates}\}$, with $u \to 0$ at infinity and Dirichlet conditions:

$$u = V_1 \quad \text{on the upper plate}, \qquad u = V_2 \quad \text{on the lower plate}. \tag{41}$$

We introduce an artificial closed boundary Γ (e.g. a rectangle containing both plates) that splits the domain into:

- Ω_{int}: the bounded interior region, treated with FEM,
- $\Omega_{\text{ext}} = \mathbb{R}^2 \setminus \overline{\Omega}_{\text{int}}$: the unbounded exterior region, treated with BEM.

3.1.2 Node partitioning

We partition the degrees of freedom into three groups:

- u_D: nodes on the plates, where the potential is **known**: $u_D = V_1$ or $u_D = V_2$,
- u_F: free interior nodes, where the potential is **unknown**,
- u_B: nodes on Γ, where the potential is **unknown** but constrained by the BEM.

The union of nodes F and nodes D are the nodes I.

3.1.3 Full block system

The FEM formulation (16) written in block form over the three groups is:

$$\begin{bmatrix} S_{FF} & S_{FD} & S_{FB} \\ S_{DF} & S_{DD} & S_{DB} \\ S_{BF} & S_{BD} & S_{BB} \end{bmatrix} \begin{bmatrix} u_F \\ u_D \\ u_B \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ T_{BB}\, G^{-1} H \end{bmatrix} = 0 \tag{42}$$

where T_{BB} acts only on Γ-nodes because interior basis functions vanish on Γ, and the plates lie inside Ω_{int} and therefore have no contribution to T.

3.1.4 Reduction of the system

The row D of (42) reads:

$$S_{DF}\, u_F + S_{DD}\, u_D + S_{DB}\, u_B = 0. \tag{43}$$

Since u_D is already known by the Dirichlet condition, this equation is **not used to solve the system**. It will be recovered a posteriori to compute the surface charge density on the plates (see Section 3.1.5).

We retain only rows F and B. Since u_D is known, we move it to the right-hand side. The reduced system is:

$$\begin{bmatrix} S_{FF} & S_{FB} \\ S_{BF} & S_{BB} - T_{BB}\, G^{-1} H \end{bmatrix} \begin{bmatrix} u_F \\ u_B \end{bmatrix} = - \begin{bmatrix} S_{FD} \\ S_{BD} \end{bmatrix} u_D, \tag{44}$$

where u_D is the vector of known Dirichlet values, with entries equal to V_1 or V_2 depending on which plate the node belongs to.

3.1.5 Recovery of the full solution

Once (44) is solved, the complete solution is recovered in the following steps.

Step 1: Potential in Ω_{int}. The FEM solution is fully determined: u_F from the solved system, u_D from the Dirichlet data, and u_B from the solved system. The potential at any interior point is:

$$u_h(x) = \sum_j u_j\, v_j(x), \qquad x \in \Omega_{\text{int}}. \tag{45}$$

Step 2: Normal derivative on Γ. The outward normal derivative on Γ is recovered from the BEM relation:

$$\frac{\partial u_B}{\partial n} = G^{-1} H \, u_B. \tag{46}$$

Step 3: Potential in Ω_{ext}. With u_B and $\frac{\partial u_B}{\partial n}$ known on Γ, the potential at any exterior point $y \in \Omega_{\text{ext}}$ is recovered via the BEM representation formula (19):

$$u(y) = \int_\Gamma G(x,y) \, \frac{\partial u_B}{\partial n}(x) \, ds_x \; - \int_\Gamma \frac{\partial G}{\partial n_x}(x,y) \, u_B(x) \, ds_x, \qquad y \in \Omega_{\text{ext}}, \tag{47}$$

which in discrete form is a matrix-vector product using the same BEM matrices G and H.

Step 4: Surface charge density on the plates. We now use the row D that was set aside in (43). With u_F, u_D, and u_B all known, it gives:

$$(S_{DF} \, u_F + S_{DD} \, u_D + S_{DB} \, u_B)_i = \int_{\text{plates}} \frac{\partial u}{\partial n} \, v_i \, ds, \qquad \forall i \in D. \tag{48}$$

This is the projection of the normal derivative onto the basis functions supported on the plates. The surface charge density is then:

$$\sigma = \varepsilon_0 \, \frac{\partial u}{\partial n}\bigg|_{\text{plates}}, \tag{49}$$

and the total charge per unit length on each plate is obtained by integrating σ along the plate.

4 NUMERICAL EXPERIMENTS

To validate the FEM-BEM coupled model, we consider a two-dimensional parallel-plate capacitor as a benchmark problem. The geometry consists of two infinitely thin conducting plates of length dx separated by a gap dy, embedded in free space ($\varepsilon_r = 1$). Dirichlet boundary conditions $u = V_1$ and $u = V_2$ are imposed on the upper and lower plates respectively, while the artificial coupling boundary Γ is placed at a distance of approximately $3\,dy$ from the plates on all sides. This choice is motivated by the spatial extent of the fringing fields: the field lines that curve around the plate edges decay on a length scale of order dy, so a margin of $3\,dy$ ensures that most of the fringing region is contained within Ω_{int} and adequately resolved by the FEM mesh. Placing Γ closer to the plates would force the FEM to resolve steep field gradients on a coarse mesh, degrading accuracy; placing it farther would increase the size of Ω_{int} and the associated computational cost without a corresponding gain in accuracy, since the BEM handles the exterior domain exactly regardless of its extent.

A key requirement for a meaningful comparison with the analytical solution is that the aspect ratio dx/dy is sufficiently large. For an ideal parallel-plate capacitor with infinite plates, the capacitance per unit length is given by the closed-form expression

$$C_{\text{ideal}} = \varepsilon_0 \varepsilon_r \frac{dx}{dy}, \tag{50}$$

which neglects fringing fields at the plate edges. When $dx/dy \geq 10$, the fringing contribution is small (typically 5–10%) and the numerical result can be meaningfully compared against (50). For this reason we adopt $dx = 0.10$ m, $dy = 0.01$ m, giving $dx/dy = 10$ and $C_{\text{ideal}} = 88.5$ pF/m. The computational domain is set to $L_x = dx + 2 \times 3\,dy$ and $L_y = dy + 2 \times 3\,dy$, and the mesh step is $\delta = 0.005$ m, yielding approximately 500 nodes and 90 nodes on Γ or $\delta = 0.002$ m, yielding approximately 2900 nodes and 234 nodes on Γ.

Several quantities are monitored to assess the correctness of the coupled solution.

Electric potential u. The potential must be monotone between the plates and lie strictly in the interval $[V_2, V_1]$ at every interior node. Outside the plates it should decay smoothly toward Γ, with no artificial oscillations near the coupling boundary. A contour plot of u is used to verify these properties visually.

Electric field $\mathbf{E} = -\nabla u$. Between the plates, and away from the edges, the field should be nearly uniform and directed perpendicularly to the plates, with magnitude close to $(V_1 - V_2)/dy$. Near the plate edges a field concentration is physically expected — the field intensity diverges at the tips of infinitely thin conductors — so a small number of elements with anomalously large $|\mathbf{E}|$ is not a sign of error but an inherent feature of the geometry.

Capacitance C. The capacitance per unit length is computed from the energy stored in the electric field as

$$C = \frac{1}{(V_1 - V_2)^2} \int_{\Omega_{\text{int}}} \varepsilon_0 |\nabla u|^2 \, d\Omega, \tag{51}$$

discretized as a sum over finite elements. Elements near the plate edges are affected by the singularity and tend to overestimate the local energy; to obtain a stable estimate, elements with $|\mathbf{E}|$ exceeding a reasonable threshold are excluded from the sum. The resulting value is expected to slightly exceed C_{ideal} due to the fringing field contribution, with a discrepancy of 5–10% for the chosen aspect ratio.

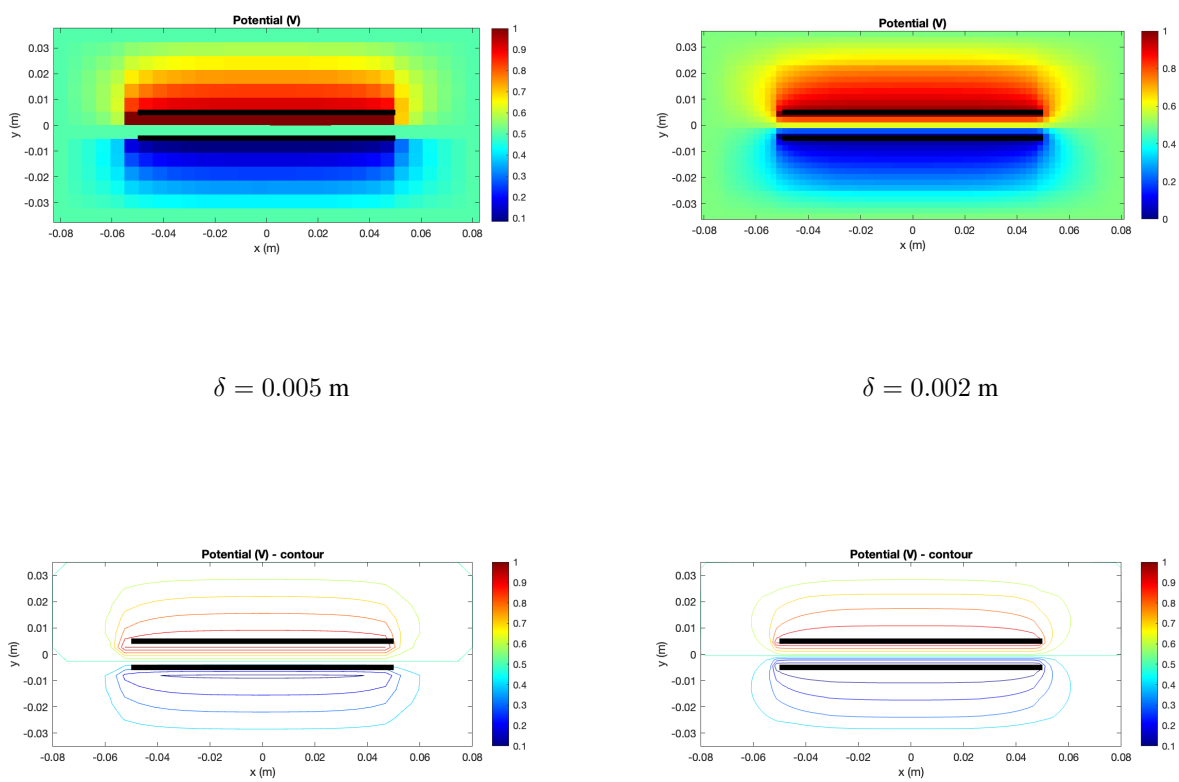
Normal derivative on Γ. The quantity $q = \partial u/\partial n$ on Γ, recovered as $q = G^{-1} H \, u_B$, represents the flux leaving Ω_{int} through the artificial boundary. Since no sources are present in Ω_{int}, the total flux through Γ must satisfy Gauss's law: the net integral of q over Γ should be zero (equal and opposite charges on the two plates). A non-zero net flux would indicate an inconsistency in the coupling.

Surface charge density on the plates. The projection of $\partial u/\partial n$ onto the plate nodes, recovered from the discarded row D of the FEM system as described in Section 3.1.5, gives the distribution of surface charge along each plate. For the ideal capacitor this distribution should be approximately uniform in the central region and exhibit concentration near the edges — a profile that is well known analytically and provides a further qualitative check on the solution.

4.1 Validation Results

The coupled FEM-BEM solver is tested on the parallel-plate capacitor described above ($dx = 0.10$ m, $dy = 0.01$ m, $dx/dy = 10$, $V_1 = 1$ V, $V_2 = 0$ V) using two mesh refinement levels: a coarse mesh with $\delta = 0.005$ m and a fine mesh with $\delta = 0.002$ m. The reference value for the capacitance per unit length, neglecting fringing fields, is $C_{\text{ideal}} = \varepsilon_0\, dx/dy = 88.5$ pF/m.

Electric potential. Figures 1 show the electric potential distribution for the two mesh levels. In both cases the potential is monotone between the plates, ranging from $V_2 = 0$ V at the lower plate to $V_1 = 1$ V at the upper plate, and decays smoothly toward Γ with no visible oscillations. The equipotential lines are horizontal and uniformly spaced in the central region between the plates, confirming the expected uniform-field behaviour, and curve regularly outward near the plate edges — the characteristic signature of fringing fields. The coarse mesh ($\delta = 0.005$ m) reproduces the overall potential distribution correctly but with noticeably lower resolution: the equipotential lines appear jagged and the field curvature at the edges is poorly resolved. The fine mesh ($\delta = 0.002$ m) produces a significantly smoother solution, capturing both the uniform region and the fringing zone with greater fidelity. This improvement is expected, as the finer discretization provides more degrees of freedom in the regions where the potential gradient changes rapidly.



$\delta = 0.005$ m $\delta = 0.002$ m

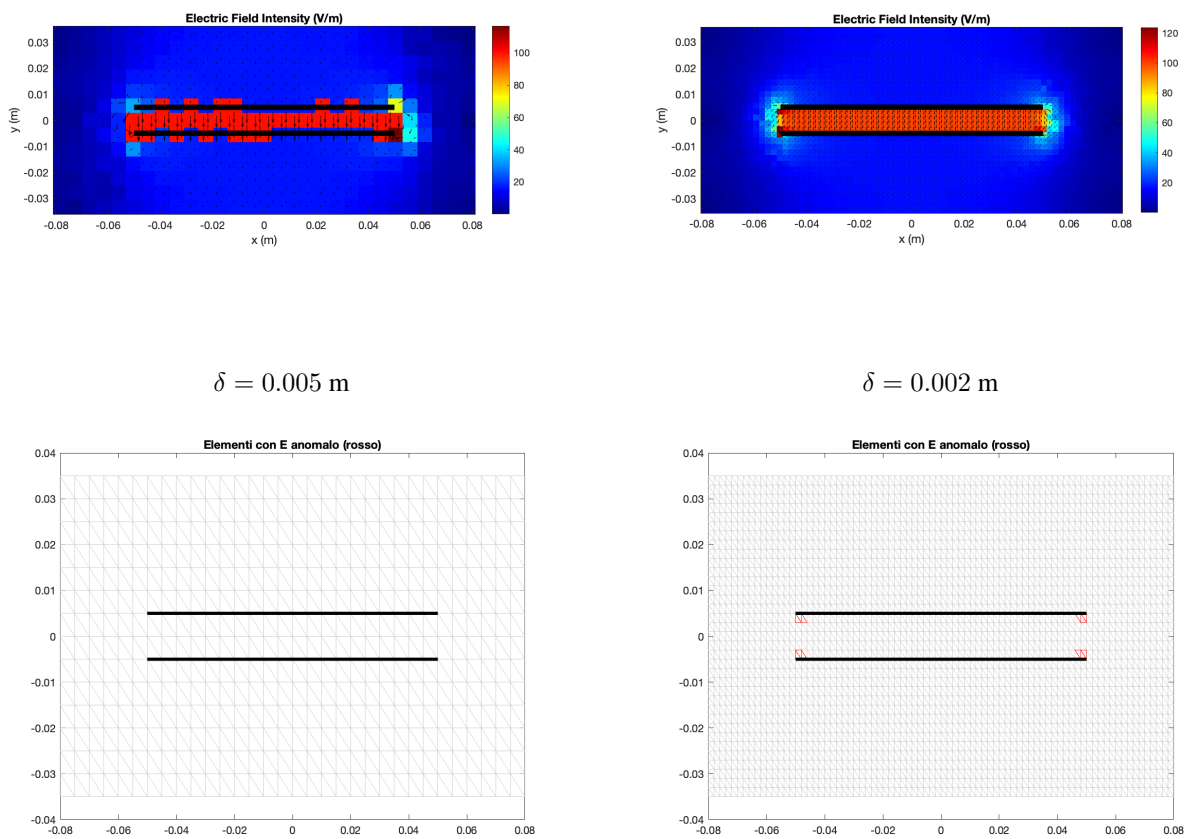
$\delta = 0.005$ m $\delta = 0.002$ m

Figure 1: Electric potential distribution (top) and equipotential lines (bottom) for two mesh resolutions.

Electric field. Figures 2 show the electric field magnitude and direction for the two mesh levels. In both cases the field is directed vertically between the plates, as expected, with magnitude close to the theoretical value $|\mathbf{E}| = (V_1 - V_2)/dy = 100$ V/m in the central region.

The coarse mesh ($\delta = 0.005$ m) resolves the field poorly: individual elements show significant variability, with values alternating above and below the expected uniform level, reflecting the inability of the coarse discretization to capture the smooth field gradient accurately. Notably, no element in this case exceeds the threshold $E_{\text{soglia}} = 1.05 \times E_{\text{teorico}} = 105$ V/m, meaning that the singularity at the plate edges is not resolved at all at this resolution — the mesh is too coarse to capture the field concentration, and the anomalous elements are simply smeared into their neighbours.

The fine mesh ($\delta = 0.002$ m) produces a markedly better result: the field is visibly uniform across the central region between the plates, and the quiver plot confirms the vertical orientation of $\mathbf{E}$ away from the edges. A small number of elements near the plate corners exceed the threshold, and their spatial distribution matches precisely the physically expected pattern — field concentration at the tips of infinitely thin conductors is a well-known singularity of this geometry and is not an artefact of the numerical method. The fact that these anomalous elements appear only with the finer mesh, and are localised at the corners, confirms that the coupled solver is behaving correctly: the finer discretization resolves the singularity rather than hiding it.



$\delta = 0.005$ m $\quad$ $\delta = 0.002$ m

$\delta = 0.005$ m $\quad$ $\delta = 0.002$ m

Figure 2: Electric field intensity and direction (top row). Elements with $|\mathbf{E}| > 105$ V/m ($> 5\%$ of $|\mathbf{E}|$)highlighted in red (bottom row).

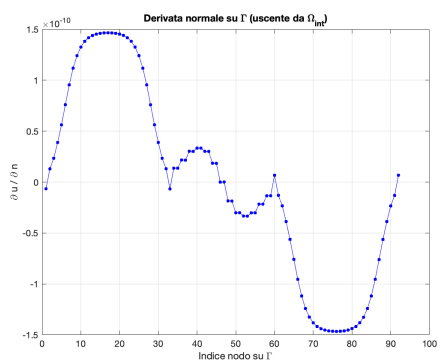
Normal derivative on Γ. Figures 3 show the normal derivative $\partial u/\partial n$ on Γ for the two mesh levels. Recalling that $\partial u/\partial n = \mathbf{E} \cdot \mathbf{n}$, this quantity represents the normal component of the electric field on the artificial boundary, i.e. the flux per unit length leaving Ω_{int} through Γ.

From a physical standpoint, the field lines connecting the two plates are contained within Ω_{int} and do not cross Γ; only the fringing fields, which curl around the plate edges and extend outward, contribute to the flux through Γ. Consequently, $\partial u/\partial n$ is expected to be negligible on the lateral sides of Γ, which are far from the plates, and to take small but non-zero values on the top and bottom sides, in correspondence with the fringing field region above and below the plates.

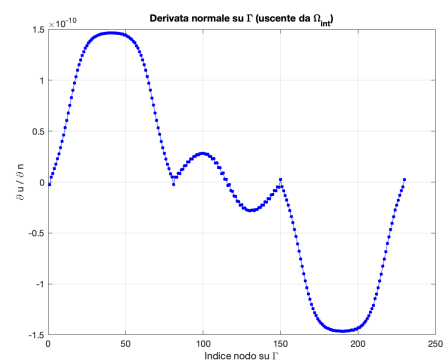
A fundamental consistency check is provided by Gauss's law: since the two plates carry equal and opposite charges, the total enclosed charge is zero and the net flux through Γ must vanish:

$$\oint_\Gamma \frac{\partial u}{\partial n}\, ds = \sum_i q_i \approx 0. \tag{52}$$

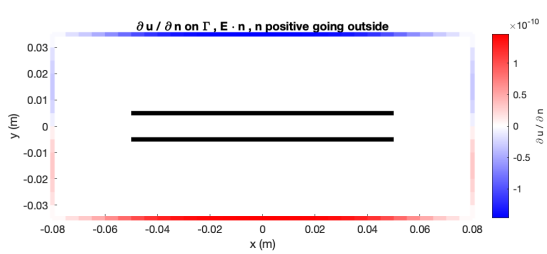
The numerical values obtained are $\sum_i q_i = -3.24 \times 10^{-13}$ for the coarse mesh ($\delta = 0.005$ m) and $\sum_i q_i = -1.43 \times 10^{-12}$ for the fine mesh ($\delta = 0.002$ m). Both values are negligibly small in absolute terms, confirming that the coupling condition is satisfied and the net flux is numerically consistent with zero.



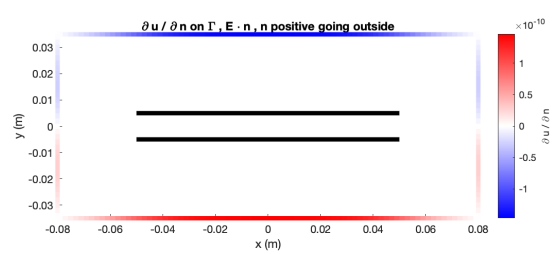
$\delta = 0.005$ m



$\delta = 0.002$ m



$\delta = 0.005$ m



$\delta = 0.002$ m

Figure 3: Normal derivative $\partial u/\partial n$ on Γ. Top row: numerical value, FEM ordered. Bottom row: visual view BEM ordered

Surface charge density. Figures 4 show the projected surface charge density on the two plates for both mesh levels. In both cases the distribution is approximately uniform in the central region of the plates and increases toward the edges, consistent with the well-known behaviour of a finite parallel-plate capacitor where field concentration at the tips produces a higher local charge density.

The two plates carry distributions of opposite sign, as required by the problem setup. The total charge per unit length on each plate is recovered by exploiting the partition of unity property of the FEM basis functions, which gives:

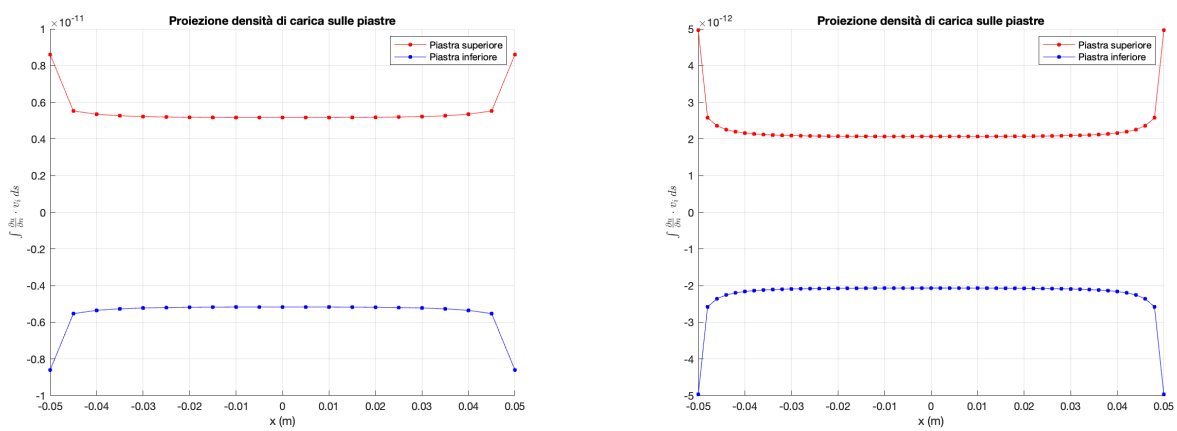
$$Q = \sum_i \left(S_{DF}\, u_F + S_{DD}\, u_D + S_{DB}\, u_B\right)_i \,, \tag{53}$$

where the sum runs over the nodes of the plate. The numerical values are reported in Table 1.

Mesh	Q_{top} (C/m)	Q_{bot} (C/m)	$Q_{\text{top}} + Q_{\text{bot}}$ (C/m)
Coarse ($\delta = 0.005$ m)	1.1687×10^{-10}	-1.1687×10^{-10}	~ 0
Fine ($\delta = 0.002$ m)	1.1426×10^{-10}	-1.1425×10^{-10}	~ 0

Table 1: Total charge per unit length on each plate. The near-perfect cancellation $Q_{\text{top}} + Q_{\text{bot}} \approx 0$ confirms global charge neutrality and the consistency of the coupled solution.

In both cases the charges on the two plates are equal and opposite to within numerical precision, confirming global charge neutrality. The fine mesh yields a slightly lower value of $|Q|$, which is consistent with the improved resolution of the potential gradient near the plate edges: the coarse mesh overestimates the local field intensity in those regions, leading to a small overestimate of the total charge. The reference value from the capacitance formula is $Q = C_{\text{ideal}} \cdot (V_1 - V_2) = 88.5 \times 10^{-12}$ C/m; the numerical values exceed this by approximately 29%, which is attributable to the fringing field contribution that is not captured by the ideal formula but is fully resolved by the coupled FEM-BEM model.



$\delta = 0.005$ m $\qquad\qquad$ $\delta = 0.002$ m

Figure 4: Projected surface charge density on the plates.

Capacitance Table 2 summarizes the capacitance values obtained with the two mesh levels. C_{full} includes all elements; C_{clean} excludes elements with $|\mathbf{E}| > 105$ V/m near the plate edges.

4.2 Comparison with Pure FEM

The coupled FEM–BEM formulation is designed to handle the unbounded exterior domain exactly, without introducing artificial truncation conditions. To assess the practical advantage of this approach, we compare

Mesh	δ (m)	C_{full} (pF/m)	C_{inside} (pF/m)	C_{clean} (pF/m)	Error w.r.t. C_{ideal}
Coarse	0.005	116.8737	88.5	88.5	32%
Fine	0.002	114.2539	88.5716	86.0078	29%
Ideal	–	88.5	88.5	88.5	–

Table 2: Capacitance per unit length for the two mesh levels.

the coupled solution against a pure FEM solver applied to the same problem on a truncated domain. In the pure FEM setting, the artificial boundary Γ is treated as a Dirichlet boundary with $u = 0$, which is an approximation that becomes more accurate as Γ is moved farther from the plates.

Two aspects are investigated: the accuracy of the pure FEM solution as a function of the domain size, and the computational cost. The goal is twofold: first, to verify that the FEM–BEM method is functioning correctly by comparing it against a well-resolved pure FEM solution (which serves as a benchmark when computed on a sufficiently large domain); second, to determine whether the coupled approach offers a meaningful advantage in terms of accuracy and computational efficiency.

Accuracy as a function of domain size. The margin between the capacitor plates and the artificial boundary Γ is varied as $3\,dy$, $5\,dy$, $10\,dy$, $20\,dy$, $40\,dy$ and $80\,dy$ where dy is the vertical spacing between the plates. The mesh step δ is kept fixed at two values: $\delta = 0.002$ and $\delta = 0.005$. For each configuration, the L^2 error on the potential is computed. The FEM–BEM solution is computed on a fixed small domain with margin $3\,dy$ and serves as the reference solution.

The L^2 error between the pure FEM solution and the FEM–BEM reference is computed on the nodes belonging to the interior region Ω_{int} common to both discretizations as:

$$\varepsilon_{L^2} = \sqrt{\frac{\sum_{i\in\mathcal{N}_c}\left(u_i^{\text{FEM}} - u_i^{\text{FEM–BEM}}\right)^2}{\sum_{i\in\mathcal{N}_c}\left(u_i^{\text{FEM–BEM}}\right)^2}} \times 100\%, \tag{54}$$

where $\mathcal{N}_c$ denotes the set of nodes in the common region, u_i^{FEM} is the pure FEM solution, and $u_i^{\text{FEM–BEM}}$ is the coupled solution interpolated at the same locations. The error is expressed as a percentage.

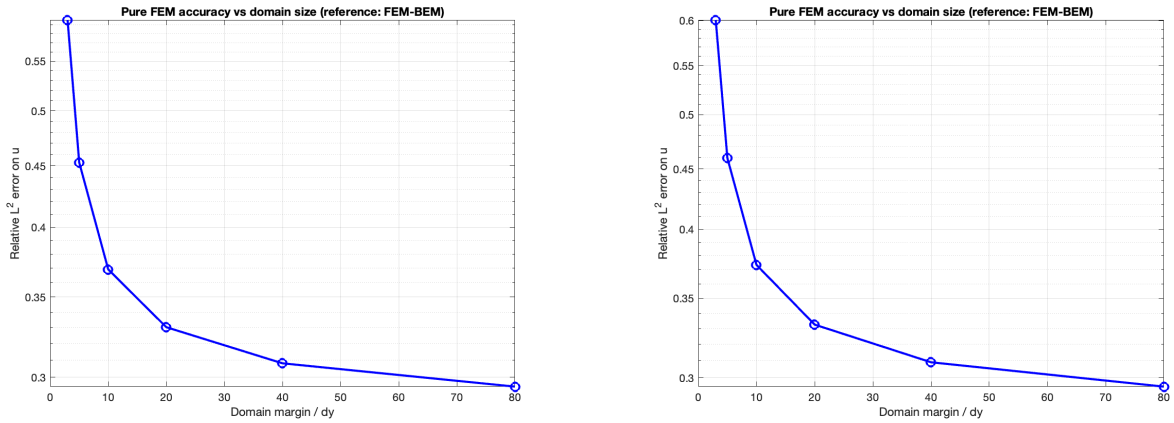
Computational cost. The wall-clock time of each simulation is measured for both the pure FEM and FEM–BEM approaches. This allows us to assess the trade-off between domain size, accuracy, and computational effort. In principle, the FEM–BEM method should achieve a given accuracy at lower cost, since it does not require enlarging the domain to reduce truncation error.

Results. The L^2 errors and execution times for both mesh resolutions ($\delta = 0.002$ and $\delta = 0.005$) are shown in Figures 5 and 6.

The results reveal some unexpected features. The L^2 error between the FEM–BEM solution and the pure FEM solution remains relatively high across all tested domain sizes, with values around 30% even when the pure FEM domain is significantly enlarged. This is substantially higher than anticipated and raises questions about whether the coupled method is converging to the correct solution, or whether there are implementation issues that need to be addressed.

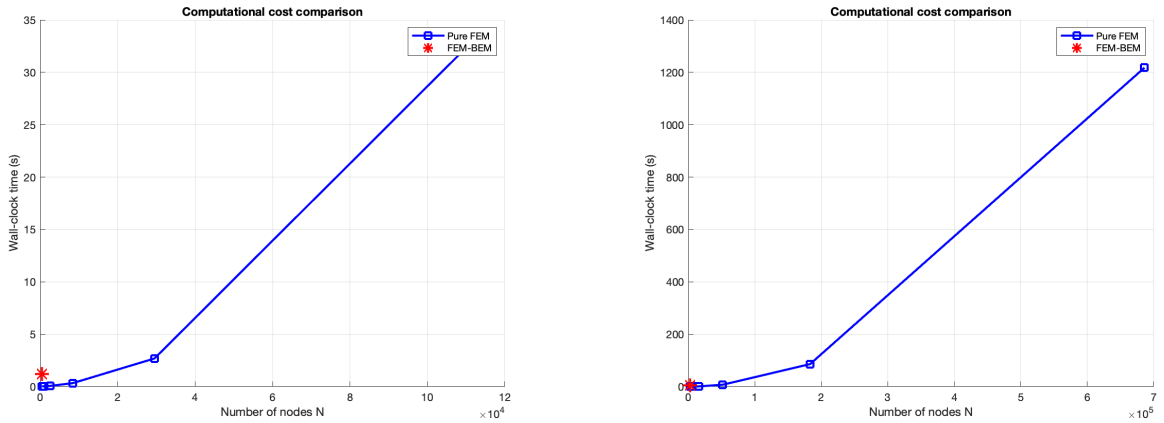
Regarding computational cost, the FEM–BEM approach does show promising execution times compared to pure FEM on larger domains. However, given that the L^2 error does not decrease toward zero as the pure FEM domain is enlarged, a direct comparison of execution times is of limited value. Without a clear convergence of the pure FEM solution to the FEM–BEM reference (or vice versa), it is difficult to establish a meaningful benchmark that would demonstrate the true computational advantage of the coupled method.

Discussion. These preliminary results suggest that further investigation is needed. The persistent L^2 error of approximately 30% indicates either: (i) a fundamental discrepancy between the two methods that requires deeper theoretical analysis, (ii) numerical issues in the implementation of the FEM–BEM coupling (such as inaccuracies in the BEM matrix assembly, quadrature errors, or incorrect handling of boundary



$\delta = 0.005$ $\delta = 0.002$

Figure 5: L^2 error (in percentage) between pure FEM and FEM–BEM solutions as a function of domain margin.



$\delta = 0.005$ $\delta = 0.002$

Figure 6: Wall-clock execution time as a function of domain margin, for pure FEM and FEM–BEM.

conditions), (iii) an inappropriately chosen reference solution or (iv) a more mesh more fine with $\delta < 0.002$.

In the following section, we will examine potential sources of this error and discuss whether the observed behavior is physically reasonable or points to a need for methodological refinement. For now, we report these results as-is and acknowledge the need for further validation before drawing definitive conclusions about the effectiveness of the FEM–BEM coupling for this problem.

5 FURTHER INVESTIGATION

In order to better understand the observed issues and to provide a clearer and more rigorous interpretation of the results, it is useful to investigate the following aspects:

- Electromagnetic theory concerning edge effects in capacitors, with particular reference to available analytical results and findings reported in the literature.
- The actual decay rate of the electric potential in the exterior domain, and whether it behaves at least

as $1/r^2$, as commonly assumed. This point should be clarified through both theoretical studies and experimental evidence available in the literature.

- The validity of solving the BEM problem under the assumption that the artificial boundary Γ is smooth, despite the presence of four geometric corners. In particular, at the corners the solid angle factor is $1/4$ rather than $1/2$, and a rigorous justification of the adopted treatment is required.
- Theoretical considerations on the appropriate choice of the domain margin size and on the ratio dy/dx, possibly guided by similar numerical experiments reported in the literature.
- The use of higher-order elements in regions where the solution exhibits stronger gradients, especially near critical points. In particular, replacing linear elements with quadratic ones may improve the local accuracy.
- The relevance of evaluating the error over the entire computational domain, especially near the edges. It may be more informative to analyze the error separately in different subregions of the domain.
- The suitability of the L^2 norm as an error metric for this problem, and whether alternative norms could provide a more meaningful assessment of the solution quality.

6 CONCLUSIONS

7 ACKNOWLEDGEMENTS

This report is based on Chapter 6 of the book *Topics in Boundary Element Research* by C.A. Brebbia [1]. The implementation of the code derives, for the FEM part, from the work of Ozlem Ozgun and Mustafa Kuzuoglu [2], distributed under the GNU General Public License, and for the BEM part, from code developed during the course *Computational Electromagnetics* by Prof. Di Rienzo. Furthermore, artificial intelligence tools have been used in the preparation of this report. The full project code is available on GitHub: `https://github.com/Michelino21/FemBem_Project`.

REFERENCES

[1] C.A. Brebbia. *Topics in Boundary Element Research*. Springer, Berlin, Germany, 1990.

[2] Ozlem Ozgun and Mustafa Kuzuoglu. Fem code for 2d electromagnetics. GNU General Public License, Version 3, 2018.

A INTRODUZIONE ITALIANA

Un classico esempio di accoppiamento FEM/BEM in un problema elettrostatico è il calcolo del campo elettrico generato da un condensatore a facce piane parallele. In una simulazione di questo tipo è spesso utile visualizzare non solo il comportamento del campo all'interno del condensatore, ma anche quello all'esterno. Nei corsi di Fisica 2, il campo esterno viene di norma assunto nullo; in realtà, a causa degli effetti di bordo, esso non è trascurabile e in alcune applicazioni può avere un ruolo rilevante. Questa è la motivazione principale per analizzare in maggiore dettaglio il comportamento del campo elettrico in un problema di questo tipo.

In questo progetto considereremo il caso relativamente semplice di un condensatore a facce piane parallele (si veda la figura di riferimento). Oltre alle cariche presenti sulle armature, non saranno considerate altre sorgenti nello spazio. L'equazione differenziale da risolvere è la prima legge di Maxwell (55) nella sua forma elettrostatica semplificata (56), trattata esprimendo il campo elettrico come gradiente del potenziale. (57)

$$\nabla \cdot D = \rho \tag{55}$$

$$\nabla \cdot E = 0 \tag{56}$$

$$\Delta V = 0 \tag{57}$$

Un primo approccio ben noto consiste nell'utilizzare il metodo degli elementi finiti (FEM). Si definisce un dominio sufficientemente ampio da contenere il condensatore e si impone sul bordo un potenziale nullo, assumendo che a distanza sufficientemente grande il potenziale tenda effettivamente a zero. Questo metodo presenta tuttavia alcuni svantaggi: richiede domini molto estesi e quindi mesh di grandi dimensioni, con conseguente aumento dei tempi di calcolo. Inoltre la finezza della mesh deve essere omogenea dentro e fuori dal condensatore, anche se la qualità richiesta per ottenere un buon risultato non è la stessa nelle due regioni. Calcolare il campo esterno tramite FEM porta quindi a un significativo spreco computazionale e, in ogni caso, il potenziale non può essere rappresentato fino all'infinito, introducendo un'inevitabile approssimazione.

Il metodo BEM, invece, offre una soluzione più efficiente al problema. Riducendo di un ordine la dimensione del dominio, permette di ottenere notevoli vantaggi in termini computazionali. Per sua natura il BEM consente di calcolare con precisione la variabile incognita in qualsiasi punto interno a un dominio, risolvendo il problema solo sul suo bordo. (fig) Il BEM risulta applicabile anche a domini molto estesi o non limitati (fig.), a patto di conoscere a priori l'andamento asintotico del potenziale e che esso converga come $1/r^2$. In tali casi, il BEM riesce a rappresentare correttamente il comportamento all'infinito senza introdurre approssimazioni artificiali.

Lo scopo di questo lavoro è quindi combinare FEM e BEM per ottenere risultati più accurati a costi computazionali inferiori. L'idea è sfruttare il FEM per determinare con precisione la soluzione all'interno del condensatore e utilizzare il BEM per estenderla verso l'esterno fino all'infinito in modo esatto. Il meccanismo di accoppiamento tra i due metodi sarà descritto in dettaglio nella sezione dedicata.